Peer Response

by Ali Alzahmi - Friday, 16 May 2025, 6:07 PM

Koulthoum, you have come up with a good reflection on the transformative effects of Industry 4.0 and 5.0 in construction. I particularly liked the way you drew attention to the use of AI, IoT, and real-time data in optimising the construction processes (Chrusciak et al., 2025). Your understanding of Building Information Modelling (BIM) shows the actual understanding of how digital tools are streamlining teamwork and eliminating design errors (Deng et al., 2022).

Your Queensferry Crossing project example is an outstanding case in the real world. It is concerning that poor PMIS implementation led to scheduling conflicts, idle resources, and an overrun of a £270 million budget (Kang et al., 2024). This is consistent with my experience in construction management where the inability to synchronize the digital systems often translates to additional costs and lack of operation efficiency.

I concur with your conclusion that the digital transformation is not enough without user's training and system integration. According to Bucci et al. (2025), Industry 5.0 does not only require the implementation of technology, but also a a human-centric plan to attain resilience. You may also reflect on the idea of digital maturity by Fernandes and Costa (2024), which identifies how organisations need to balance out technological readiness against capability of workforce so as to get maximum value.

In conclusion, your post suggests realistic and pragmatic perspective of the digital transformation journey in construction. Your suggestion on investing in technology and digital literacy has strong backing and is tremendously relevant to the industry players who are interested in sustainable and effective outcomes of their projects.

References

Bucci, I., Fani, V., & Bandinelli, R. (2025). Towards human-centric manufacturing: Exploring the role of human digital twins in Industry 5.0. *Sustainability*, 17(1), 129. Available at: https://doi.org/10.3390/su17010129 (Accessed: 16 May 2025).

Chrusciak, C. B., Szejka, A. L., & Canciglieri Junior, O. (2025). Integrating digital transformation with human-centric factors strategies to enhance organisational process performance: The H.O.P.E. Model. *Journal of Industrial Information Integration*, 44, 100785. Available at: https://doi.org/10.1016/j.jii.2025.100785 (Accessed: 16 May 2025).

Deng, M., Wang, X., Li, D., et al. (2022). Digital ID framework for human-centric monitoring and control of smart buildings. *Building Simulation*, 15, 1709–1728. Available at: https://doi.org/10.1007/s12273-022-0902-3 (Accessed: 16 May 2025).

Fernandes, L. L. de A., & Costa, D. B. (2024). A conceptual model for measuring the maturity of an intelligent construction environment. *Construction Management and*

Economics, 42(12), 1403-1426. Available at: https://doi.org/10.1080/17452007.2024.2364693 (Accessed: 16 May 2025).

Kang, H., Kim, H., Hong, J., Jeoung, J., Lee, M., & Hong, T. (2024). Human-centered intelligent construction for sustainable cities. *Automation in Construction*, 168, 105788. Available at: https://doi.org/10.1016/j.autcon.2024.105788 (Accessed: 16 May 2025).

In reply to Fahad Abdallah

Peer Response

by Ali Alzahmi - Friday, 16 May 2025, 6:12 PM

Fahad, your Industry 4.0 and 5.0 review of healthcare is thoughtful and completely relevant. What I especially liked about your approach was the attention you paid to the practical manifestation of the digital innovation in the real context of healthcare. The story that you gave about the 2022 failure of the NHS Meditech Electronic Health Record (EHR) system is particularly powerful (Barmecha & Last, 2023). As you correctly pointed out, this outage had severe effects on healthcare delivery with service interruption of important patient data like their medical history, lab results, and course of treatment. This disruption caused late treatments, scheduling of the appointments, and chances of medical errors (Mondal & Sameer, 2025). From my professional experience in the support of an IT upgrade in a hospital, I have found similar problems when temporary systems' downtimes impacted patient care, triggered staff frustration and workflow interruptions.

Regarding the financial and reputational implications of such a failure (Stanimirovic, 2024), the author discusses the need to build resilient digital infrastructures in healthcare. Further on, Stanimirovic (2024) state that the human-centric principles of Industry 5.0 (such as safety, resilience, trust, etc.) need to inform healthcare technological advancement in order to ensure safety to both patients and service providers.

I completely concur with your recommendation of necessity of investment to go beyond upgradation to workforce training, backup systems and overall precautionary measures on risks. Coelho et al. (2023) confirm this by highlighting the need for organisations to merge technological advancement with human resilience approach. You may also refer to the findings of Osipov, (2021), where the author identifies the relevance of system redundancy and user preparedness to maintain the provision of healthcare services in a safe mode even in cases of technological failure.

All in all, your analysis provides a balanced and realistic view of the situation. Your conclusion that the healthcare organisations should equally consider the aspects of technological efficiency and patient safety reflects the true sentiment of Industry 5.0 principles.

References

Barmecha, J., & Last, Z. (2023). Electronic health record and patient safety. In A. Agrawal & J. Bhatt (Eds.), *Patient safety* (Chapter 9). Springer, Cham. Available at: https://doi.org/10.1007/978-3-031-35933-0_9 (Accessed: 16 May 2025).

Coelho, P., Bessa, C., Landeck, J. and Silva, C. (2023) 'Industry 5.0: The Rise of a Concept', *Procedia Computer Science*, 217, pp. 1137–1144. Available at: https://doi.org/10.1016/j.procs.2022.12.312 (Accessed: 16 May 2025).

Mondal, R., & Sameer, M. (2025). Connected healthcare system technology interventions to improve patient safety by reducing medical errors: A systematic review. *Global Journal on Quality and Safety in Healthcare*, 8(1), 43–49. Available at: https://doi.org/10.36401/JQSH-24-23 (Accessed: 16 May 2025).

Osipov, V.S. and Skryl, T.V., (2021). Impact of digital technologies on the efficiency of healthcare delivery. In *IoT in healthcare and ambient assisted living* (pp. 243-261). Singapore: Springer Singapore. Available at: https://link.springer.com/chapter/10.1007/978-981-15-9897-5_12 (Accessed: 16 May 2025).

Stanimirovic, D. (2024). Failures and fallacies of eHealth initiatives: Are we finally able to overcome the underlying theoretical and practical orthodoxies? *Digital Health*. Available at: https://doi.org/10.1177/20552076241254019 (Accessed: 16 May 2025).